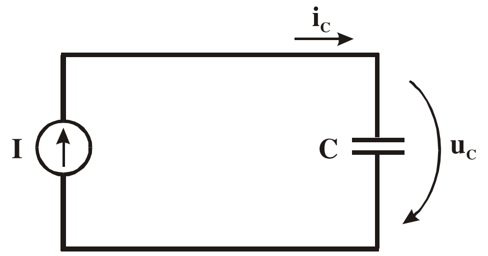
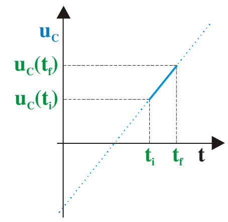
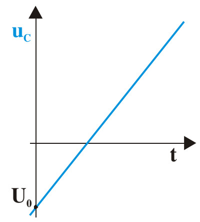
Condensador Percorrido por uma Corrente Constante



$$i_C(t) = I \quad \Rightarrow \quad \frac{d[u_C(t)]}{dt} = \frac{I}{C} \quad (V/s)$$

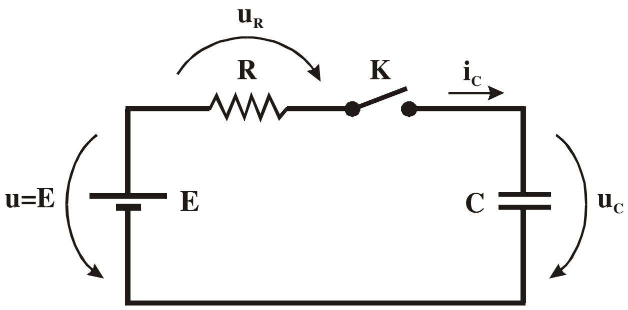




$$u_C(t_f) = \frac{I}{C}\cdot(t_f - t_i) + u_C(t_i)$$

$$u_C(t) = \frac{I}{C}\cdot t + U_0$$

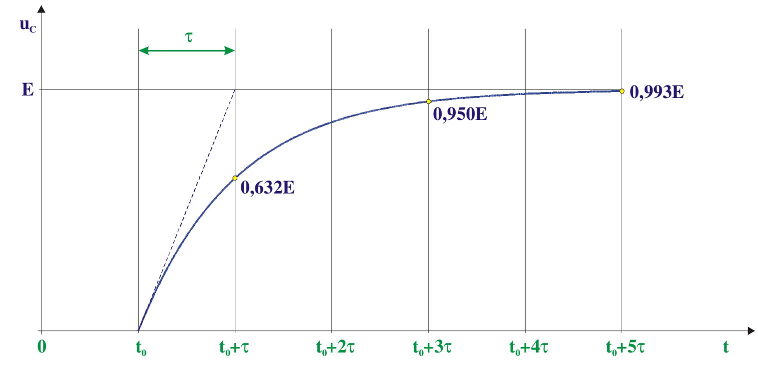
Resposta Forçada do Circuito RC de 1ª Ordem

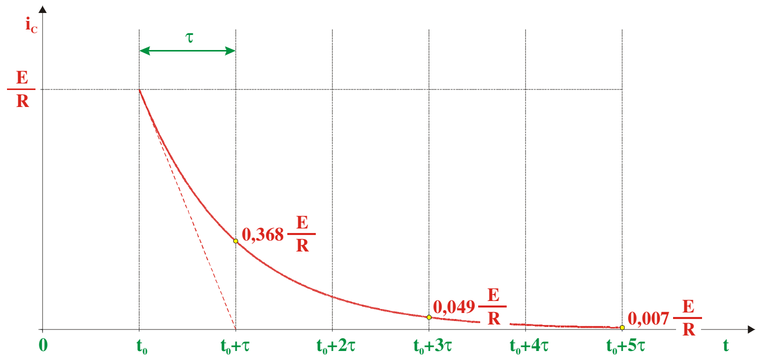


$$\begin{cases} u_C(t) = 0 \text{ em } t = t_0 \\ \text{K é fechado em } t = t_0 \end{cases}$$

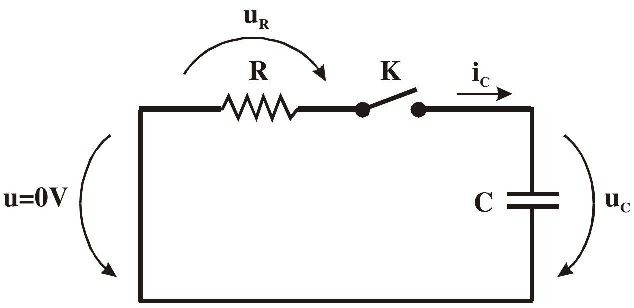
$$t \geq t_0 \quad \Rightarrow \quad \begin{cases} u_C(t) = E - E\cdot e^{-\frac{1}{RC}\cdot(t-t_0)} \\ i_C(t) = \frac{E}{R}\cdot e^{-\frac{1}{RC}\cdot(t-t_0)} \end{cases}$$

Constante de tempo do circuito: $\tau = RC$ (s)





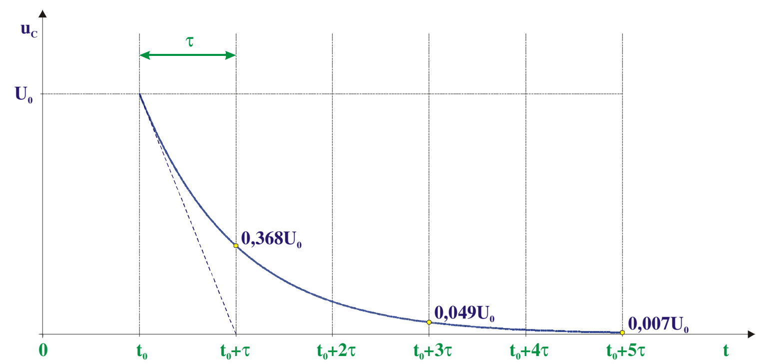
Resposta Natural do Circuito RC de 1ª Ordem

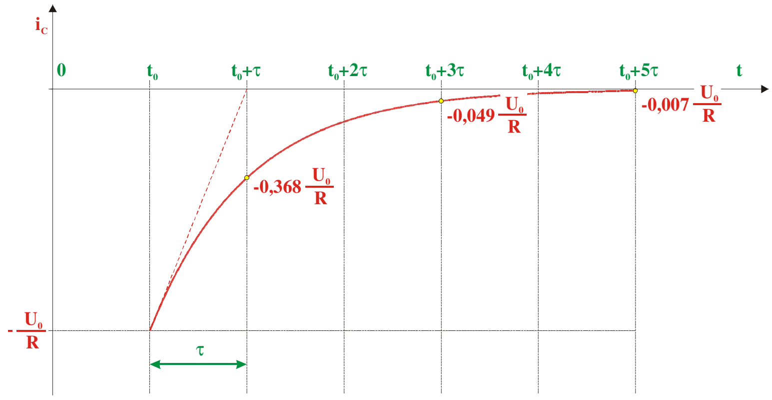


$$\begin{cases} u_C(t) = U_0 \text{ em } t = t_0 \\ \text{K é fechado em } t = t_0 \end{cases}$$

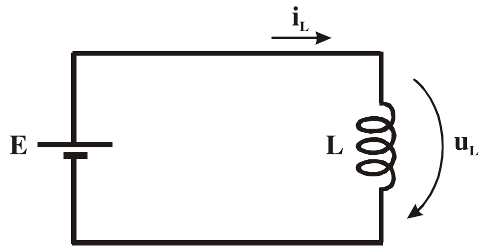
$$t \geq t_0 \quad \Rightarrow \quad \begin{cases} u_C(t) = U_0\cdot e^{-\frac{1}{RC}\cdot(t-t_0)} \\ i_C(t) = -\frac{U_0}{R}\cdot e^{-\frac{1}{RC}\cdot(t-t_0)} \end{cases}$$

Constante de tempo do circuito: $\tau = RC$ (s)

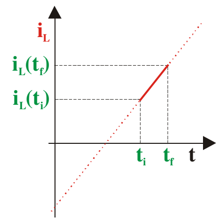


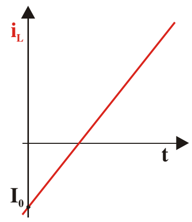


Bobina Submetida a uma Tensão Constante



$$u_L(t) = E \quad \Rightarrow \quad \frac{d[i_L(t)]}{dt} = \frac{E}{L} \quad \text{(A/s)}$$

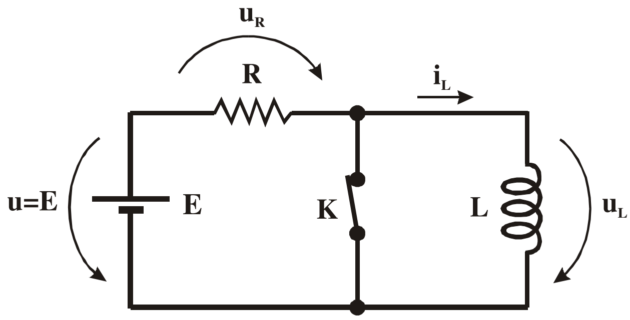


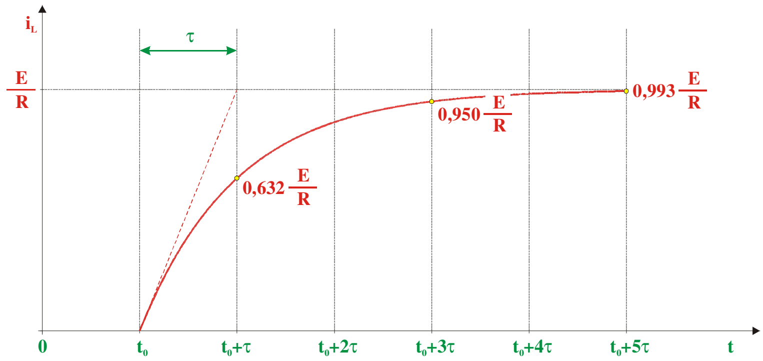


$$i_L(t_f) = \frac{E}{L}\cdot(t_f - t_i) + i_L(t_i)$$

$$i_L(t) = \frac{E}{L}\cdot t + I_0$$

Resposta Forçada do Circuito RL de 1ª Ordem

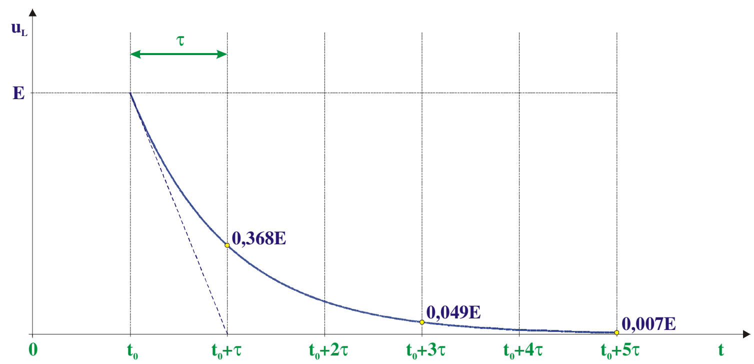




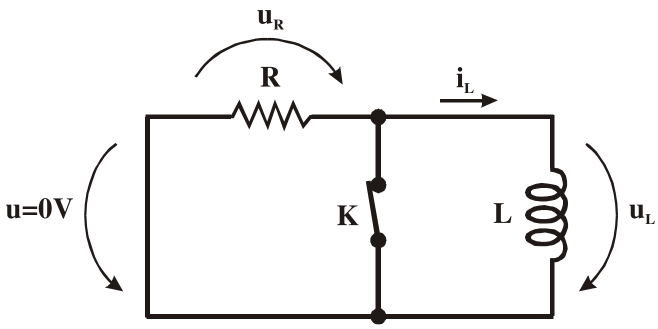
$$\begin{cases} i_L(t) = 0 \text{ em } t = t_0 \\ \text{K é aberto em } t = t_0 \end{cases}$$

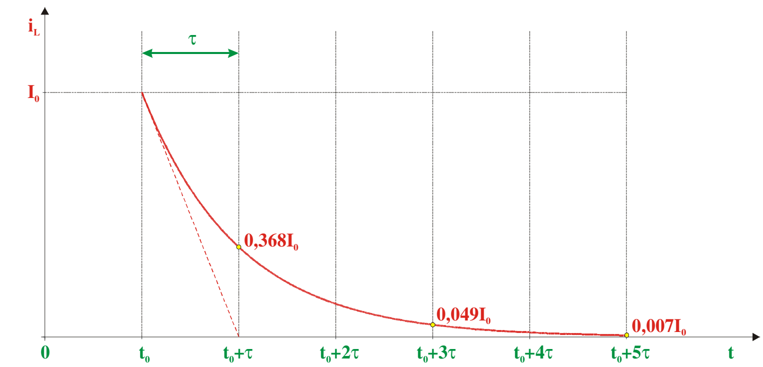
$$t \geq t_0 \quad \Rightarrow \quad \begin{cases} i_L(t) = \dfrac{E}{R} - \dfrac{E}{R}\cdot e^{-\frac{R}{L}\cdot(t-t_0)} \\ u_L(t) = E\cdot e^{-\frac{R}{L}\cdot(t-t_0)} \end{cases}$$

Constante de tempo do circuito: $\tau = \dfrac{L}{R}$ (s)



Resposta Natural do Circuito RL de 1ª Ordem





$$\begin{cases} i_L(t) = I_0 \text{ em } t = t_0 \\ \text{K é aberto em } t = t_0 \end{cases}$$

$$t \geq t_0 \quad \Rightarrow \quad \begin{cases} i_L(t) = I_0\cdot e^{-\frac{R}{L}\cdot(t-t_0)} \\ u_L(t) = -R\cdot I_0\cdot e^{-\frac{R}{L}\cdot(t-t_0)} \end{cases}$$

Constante de tempo do circuito: $\tau = \dfrac{L}{R}$ (s)